Worksheet (Number Theory, Counting & Graph Theory)

Number Theory

1.

What time does a 24-hour clock read

a) 100 hours after it reads 2:00?
b) 45 hours before it reads 12:00?
c) 168 hours after it reads 19:00?

2.

Suppose that a and b are integers, $a \equiv 4 \pmod{13}$, and $b \equiv 9 \pmod{13}$. Find the integer c with $0 \leq c \leq 12$ such that

a) $c \equiv 9a \pmod{13}$.
b) $c \equiv 11b \pmod{13}$.
c) $c \equiv a + b \pmod{13}$.
d) $c \equiv 2a + 3b \pmod{13}$.
e) $c \equiv a^2 + b^2 \pmod{13}$.
f) $c \equiv a^3 - b^3 \pmod{13}$.

3.

Suppose that a and b are integers, $a \equiv 11 \pmod{19}$, and $b \equiv 3 \pmod{19}$. Find the integer c with $0 \leq c \leq 18$ such that

a) $c \equiv 13a \pmod{19}$.
b) $c \equiv 8b \pmod{19}$.
c) $c \equiv a - b \pmod{19}$.
d) $c \equiv 7a + 3b \pmod{19}$.
e) $c \equiv 2a^2 + 3b^2 \pmod{19}$.
f) $c \equiv a^3 + 4b^3 \pmod{19}$.

4.

Find each of these values.

a) $(-133 \mathbf{~mod~} 23 + 261 \mathbf{~mod~} 23) \mathbf{~mod~} 23$
b) $(457 \mathbf{~mod~} 23 \cdot 182 \mathbf{~mod~} 23) \mathbf{~mod~} 23$

5.

Find each of these values.

a) $(99^2 \textbf{ mod } 32)^3 \textbf{ mod } 15$

b) $(3^4 \textbf{ mod } 17)^2 \textbf{ mod } 11$

c) $(19^3 \textbf{ mod } 23)^2 \textbf{ mod } 31$

d) $(89^3 \textbf{ mod } 79)^4 \textbf{ mod } 26$

6.

Decide whether each of these integers is congruent to 3 modulo 7.

a) 37 **b)** 66

c) -17 **d)** -67

Counting

1.

Suppose I roll red and blue dice!

ICP 11-2 In how many outcomes the dice show different values?

2.

ICP 11-4 How many ways to arrange the letters A, B, C, and D ?

3.

ICP 11-5 How many ways to arrange n letters ?

4.

The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

5.

How many different bit strings of length seven are there?

6.

Counting Functions How many functions are there from a set with m elements to a set with n elements?

7.

Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

8.

How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

9.

Suppose that there are eight runners in a race. The winner receives a gold medal, the second-place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties?

10.

How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a standard deck of 52 cards?

Graph Theory

1.

What are the degrees and what are the neighborhoods of the vertices in the graphs G and H displayed in Figure 1?

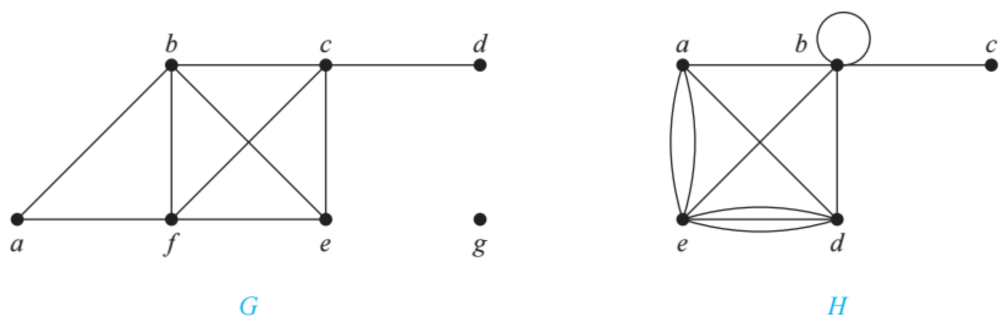
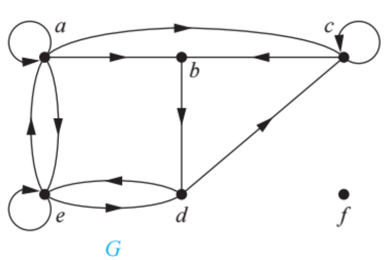


FIGURE 1 **The Undirected Graphs G and H.**

2.

Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure 2.



3.

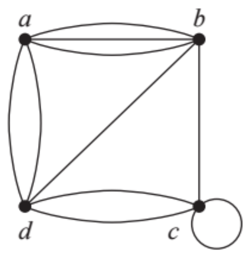
Draw a graph with the adjacency matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

with respect to the ordering of vertices a, b, c, d.

4.

Use an adjacency matrix to represent the pseudograph shown in Figure 5.



5.

Represent the graph shown in Figure 6 with an incidence matrix.

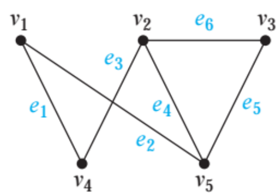


FIGURE 6 **An Undirected Graph.**

6.

Represent the pseudograph shown in Figure 7 using an incidence matrix.

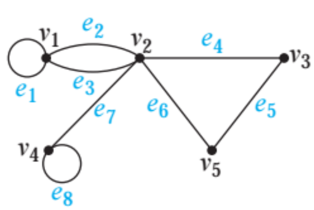


FIGURE 7
A Pseudograph.